

TEXT ANALYTICS

Course Infos

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Ph.D in Data Science from Scuola Normale Superiore

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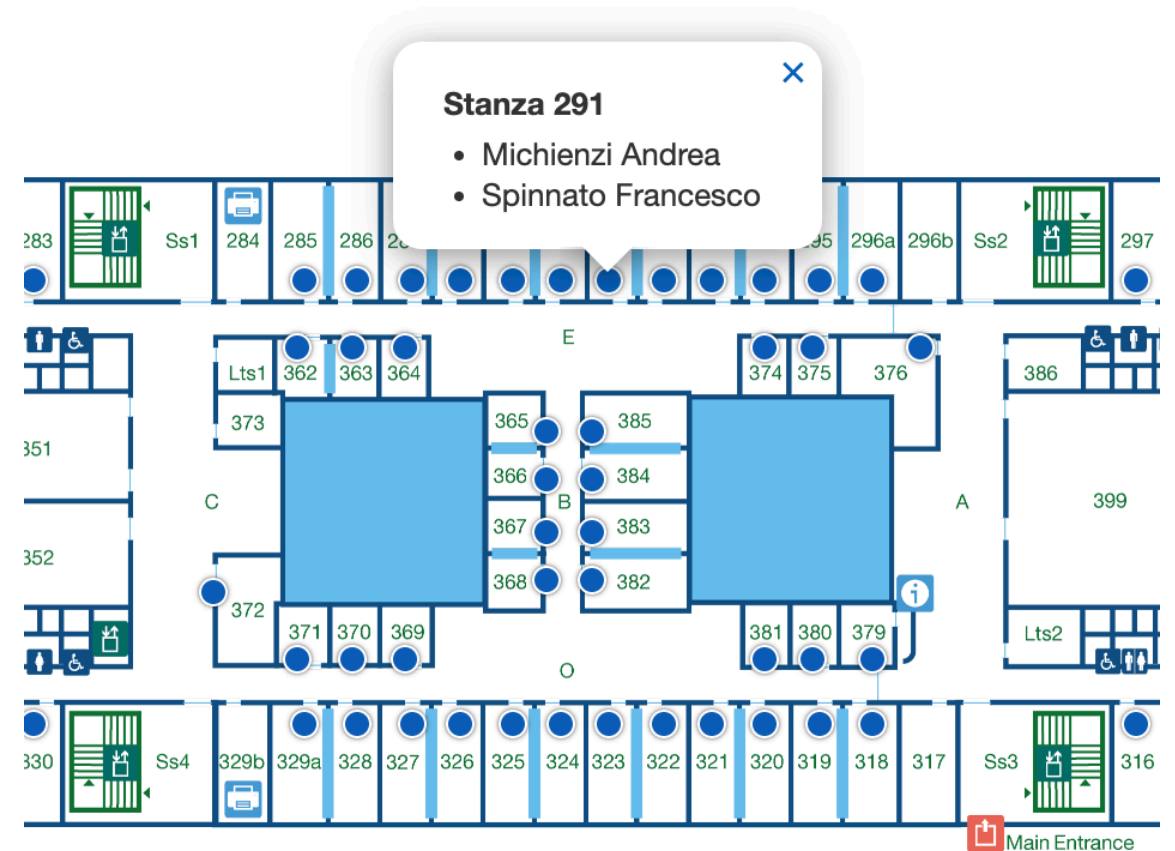
Research activity: explainable artificial intelligence for sequential data (time series, sequences, trajectories, text...)

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Web site: <https://fspinna.github.io>

Office: Room 291, Dipartimento di Informatica

Office hours: TBD, for now by appointment, send an email/teams message.



Course Info

- Title: Text Analytics
- Course code: 635AA
- Academic year: 2026/2027
- 6 CFU

Polo Fibonacci B

Day	Time	Room
Monday	11:00-13:00	M1
Friday	14:00-16:00	L1

Public Reference Webpage:

<https://fspinna.github.io/teaching/text-analytics/>

- Course information
- Lecture slides
- Articles and course materials

[Teams](#) Reference Webpage

- Announcements

Lecture Types

Frontal

- theory

Practical

- python notebooks
- you can run jupyter notebooks:
 - locally using Anaconda Navigator or Miniconda (preferred way)
 - online, e.g., Colab Notebooks



Prerequisites

Needed prerequisites:

- Basic knowledge of mathematics
- Basic knowledge of programming in **python**

Useful prerequisites:

- Probability theory
- Information theory
- Data mining and Machine learning Algorithms

Text analytics is a course that touches a lot of topics:

- "Information Retrieval": 289AA, document indexing and retrieval
- "Data mining": 420AA module 1, data mining basics
- "Machine Learning": 654AA, machine learning for NLP
- "Social network analysis", 668AA, social network advanced
- "Visual analytics", 6002AA, visual metaphors for reporting

Main Materials

Course Slides and Notebooks, Francesco Spinnato
<https://fspinna.github.io/teaching/text-analytics/>

Speech and Language Processing, Daniel Jurafsky,
<https://web.stanford.edu/~jurafsky/slp3/>

Deep Learning, I. Goodfellow, Y. Bengio and A. Courville,
<https://www.deeplearningbook.org>

Foundations of Large Language Models, Tong Xiao and Jingbo Zhu,
<https://arxiv.org/abs/2501.09223>

How to study for this course

How to read the materials:

- Slides and notebooks should be mostly sufficient.
- The slides contain formulas because they are an excellent way to summarize data, models, and concepts...
- ... but I will highlight those that are really important and you need to know.
- The most important thing is to understand the concepts,
- and have a practical understanding on how to deal with text analytics

A substantial part of the teaching material is derived from the previous editions of the Text Analytics course. Materials from previous years can be found at:

- (2023 - 2026) <https://laurapollacci.github.io/>
- (2017 - 2023) <https://didawiki.di.unipi.it/doku.php/mds/txa/>

A big thanks to the previous instructors for the materials!

Roadmap (Tentative)

September

- Intro to the course
- Text Processing and Linguistic Analysis
- Probabilistic Language Models

October

- Representing Text and Information Retrieval
- Machine Learning for Text Analytics
- Evaluation Protocols
- Data Gathering and Annotation

November

- Introduction to Neural Networks
- Neural Language Models
- Recurrent Neural Networks
- Attention and Transformers

December

- Large Language Models
- Agentic LLMs
- Applications and Seminars

Course Objectives

By the end of the course you should have:

1. **The essential techniques, algorithms and models** used in Natural Language Processing, from counting words to prompting a large language model.
2. **An understanding of how a text analytics application is put together:** which components exist, what each one assumes about its input, and how they compose into a pipeline.
3. **The ability to design, implement and evaluate** an application that analyzes, interprets or transforms text, and to say honestly how well it works.

Expected Learning Outcomes

By the end of the course you should be able to:

- **Choose** a representation for a text given a task, and justify the choice against the alternatives.
- **Implement** a working pipeline end to end: acquire text, normalize it, represent it, model it, evaluate it.
- **Evaluate** a system with the right metric, and recognize when a reported number is misleading.
- **Read** a current NLP paper or model card and place it in the landscape you have been given.
- **Decide** when a classical method suffices and when a neural or generative one earns its cost.

Exam

Exam: 2 Modalities

Written Exam + Oral Exam ANYONE

Written exam

- 2 hours (no material allowed)
- Typically 4 open questions

Oral exam

- Discussion about written exam
- Questions about course topics

Group Project + Oral Exam ONLY FOR ATTENDING STUDENTS

Group Project

- a paper (10 pag. max) + code
- a presentation

Oral exam

- Discussion about the project
- Questions about course topics

IMPORTANT: From January 2027, all students (even from previous years) are required to follow the current course guidelines and program.

Written Exam + Oral Exam

Anyone can decide to take the exam in this modality

Written exam

- 2 hours (no material allowed)
- Typically 4 open questions

Oral exam

- Discussion about written exam
- Questions / exercises about course topics

The grade of the written exam is the starting grade for the oral exam

Group Project + Oral Exam

Only **attending students** may replace the written exam with a project

- Oral exam in the first or second winter exam session (Jan-Feb 2027)
- Task: to develop a text analytics application
- Topic: a challenge (SEMEVAL, EVALITA, Kaggle), a research paper, propose your own

Final Submission

- Deadline: 7 days before the date set for a regular exam session
- Outcome:
 - paper reporting on the activity (max 10 pages)
 - Python code (with comments)
- Presentation and discussion of the project, together with individual questions on the project during the oral exam (constituting 50% of the final grade).

Group Project: FAQ

Why should I do it?

- Because you will truly learn skills that are very useful

Who can participate?

- Students who regularly attend lessons

Can we do the project at any time?

- NO, the project will be **developed during the course** and **submitted and discussed** in the first or second winter exam session (Jan-Feb 2027)
- Students **MUST** agree a project with the teacher within October 31st

What's the group composition?

- Groups (1 to 3 students)
- Group Registration form: **TBD**

What happens if the project is not sufficient?

- If it was the first winter exam session you can retry in the second session by fixing / redoing the insufficient parts
- If it was the second winter exam session you will have to do the written exam.

What happens if the oral is not sufficient?

- If it was the first winter exam session you can retry the oral in the second winter exam session, keeping the project
- If it was the second winter exam session you will have to do the written exam.

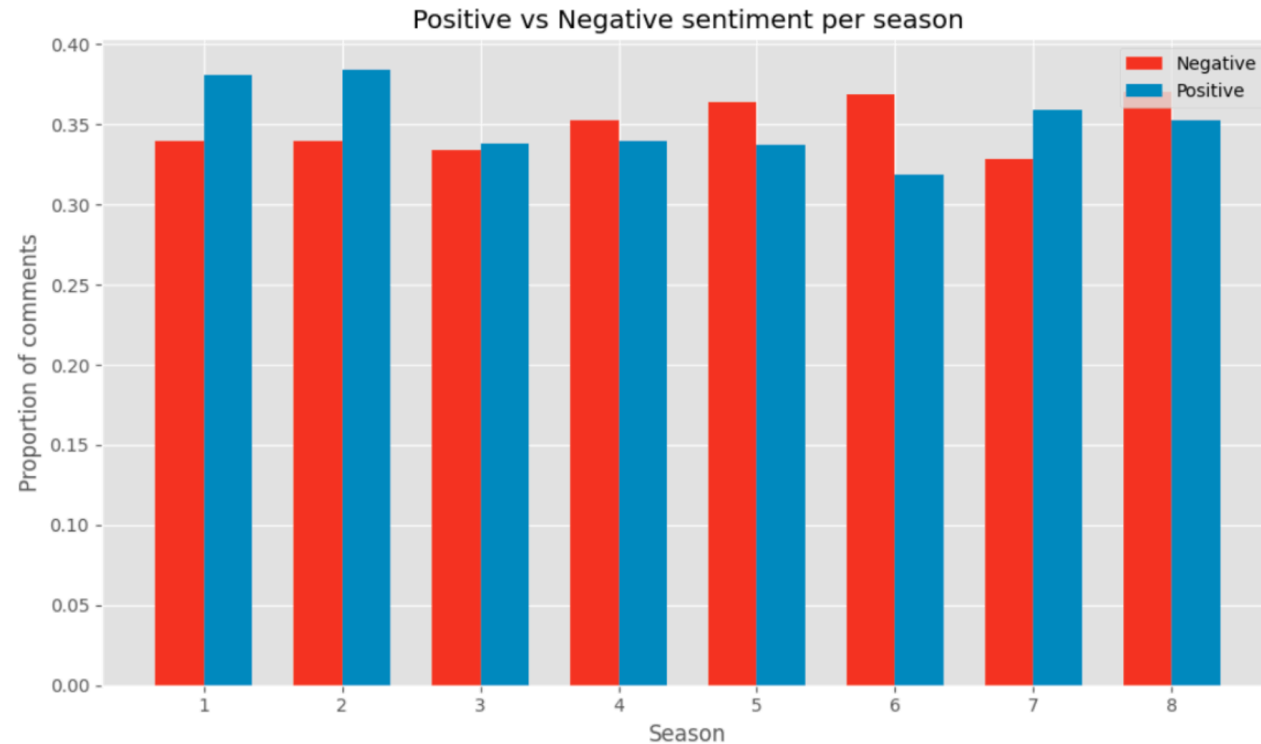
Group Project: Timeline

1. Register a team using the form: each team has a captain who takes care of communications with the teacher
2. The captain sends a project proposal to the teacher on behalf of the team
3. The proposal is agreed during office hours with the teacher (within October 31st, earlier is better)
4. Midterm update during office hours with the teacher to check how it is going (November/December)
5. Write a report with results
6. Final submission: code and report 7 days before a date set for a regular exam
7. Oral Exam: Project discussion and questions on the course topics (Jan-Feb 2027 Exam sessions)

Project Ideas: Topics

- Fake reviews detection
 - News classifications
 - Toxicity detection/Hate speech detection
 - Text classification, e.g., classification of the song genre, of the movie genre
 - Sentiment Analysis/Emotion Detection
 - Rating Predictions
 - Identify whether a text was written by a human or generated by AI
 - Research Gaps
 - Propose yours!
- Scientific NLP competitions
 - Semeval:
<https://semeval.github.io/>
 - Evalita:
<https://www.evalita.it/campaigns/evalita-2026/tasks/>
 - Desegma-IT - Detection and Segmentation of Machine Generated Text in Italian:
<https://desegma.github.io/>

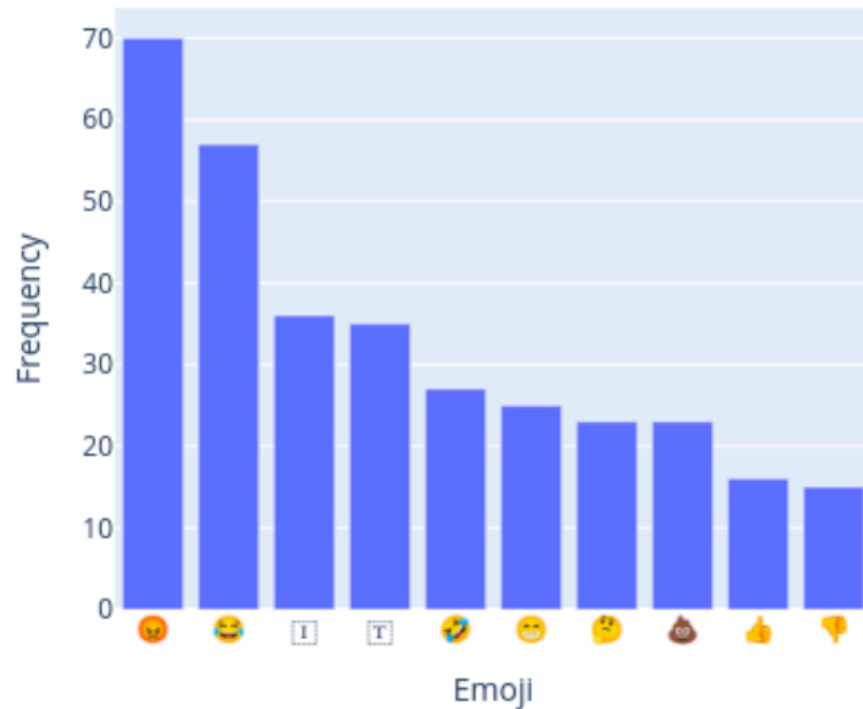
From Westeros to the Web: an analysis of Game of Thrones reviews



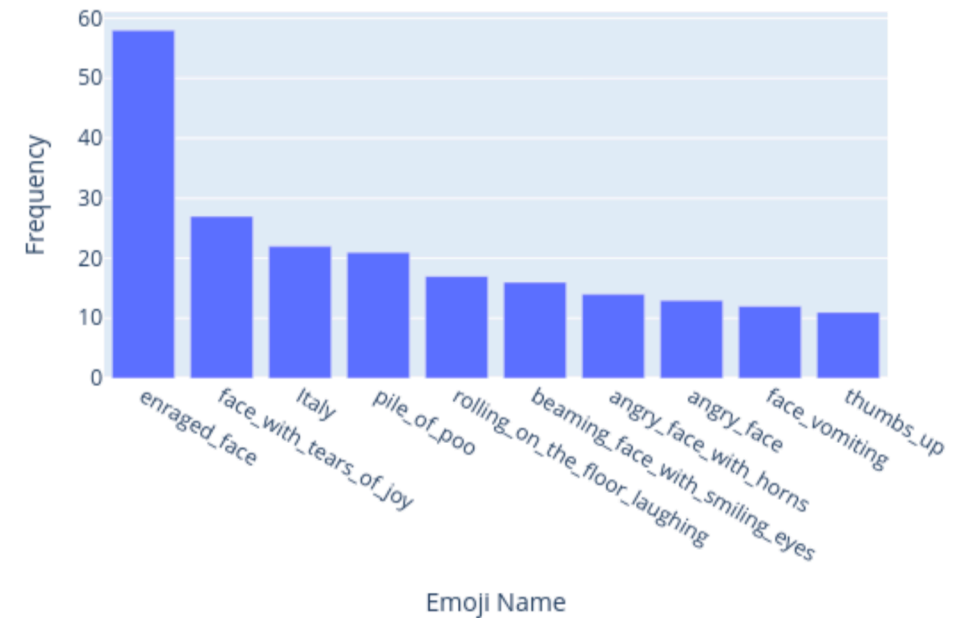
Project Ideas: Old Reports

Hate speech detection in Italian tweets

Group 4



Top 10 emojis in Hate Speech tweets



Movie Genre Classification: a Comparative Study of text Vectorization Techniques and ML Application

Genre Distribution

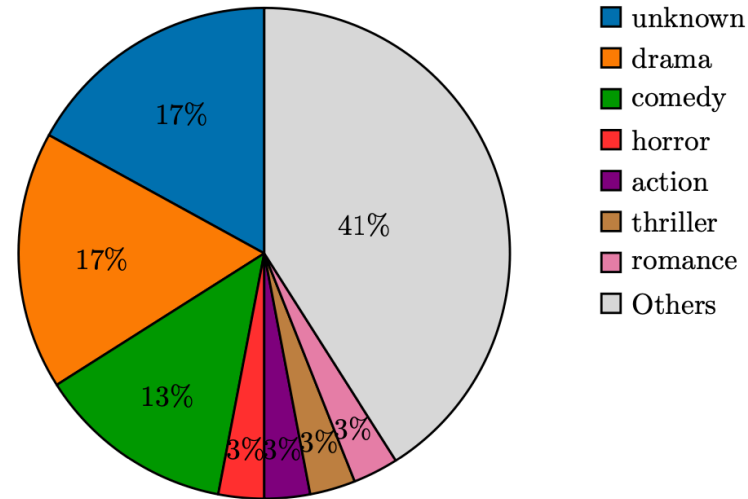


Figure 1: Proportion of the six most prominent labels under the "Genre" column, with the remaining labels categorized as "Others"

Project Ideas: Websites

Kaggle

- Many datasets, benchmarks and code
- Datasets: <https://www.kaggle.com/datasets?tags=13204-NLP>

SoBigData

- Datasets created by researchers
- Catalog: <https://sobigdata.d4science.org/group/resourcecatalogue/data-catalogue>

A proposal is a short document containing the following sections:

- Group id
- Motivation
- Goal of the project
- Available data
- Implementation and evaluation (idea/tentative)
- References

Project Proposal: Example - Group X

Motivation

- Online reviews play a very important role in influencing users' choice of whether or not to book/buy/rent etc...
- It is well-known to have fake reviews to gain/penalize credibility

Project goal

- The goal of this project is to identify patterns in the text that allow classifying it according with [a certain task] (e.g., identify polarity, misogyny, fake news...)

Available data

- Available at: kaggle, github, or other link
- Or: We build our dataset by merging DataSetA1 [1], containing N post, DataSetA2 [2], containing M post..., and scraping this website
- Context: This corpus consists of truthful and deceptive hotel reviews of 20 Chicago hotels...
- Content: 400 truthful positive reviews from X, 400 deceptive positive reviews from Mechanical Turk...

Implementation

- Text classification task (e.g., sentiment, fake, truthful vs deceptive, ...)
- Traditional ML models, CNN, RNN, Transformer
- Evaluation with Metrics: Accuracy, Precision, Recall, F1-score

References

- [1] M. Ott, et al. 2011. Finding Deceptive Opinion Spam...
- [2] ...

Group Project: Advices

- Set realistic expectations (task, computational power, ...)
- Proceed with method in the definition of the problem, the use of training, validation, and test data, and in the choice of evaluation measures. It is much more important to work with method than to achieve incredible results.
- Be dedicated. The project will require respect deadlines, attending tutorials, making calls, chats, etc. as well as multiple hours of self-educating.
- Have Patience. Be Responsive. Treat your colleagues with professionalism and respect

Questions?